THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · NCERT-TYPE MCQ TEST SET 3

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. NCERT-type: questions in the style of the textbook examples and exercises.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. The number of decimal places in the expansion of $\frac{7}{2^2 \times 5^4}$ is: 1

(a) 4 **(b)** 3 **(c)** 6 **(d)** 5

2. $(a+b)^3$ equals: 1

(a) $a^3 + 3a^2b - 3ab^2 + b^3$ **(b)** $a^3 + 3a^2b + 3ab^2 + b^3$ **(c)** $a^3 + 3ab + b^3$ **(d)** $a^3 + b^3$

3. A chord is at distance 24 cm from the centre of a circle of radius 25 cm. Its length (in cm) is: 1

(a) 14 **(b)** 16 **(c)** 13 **(d)** 15

4. A square has the same perimeter as a circle of circumference 88 cm. The side of the square (cm) is: 1

(a) 24 **(b)** 21 **(c)** 23 **(d)** 22

5. The probability of an event E always satisfies: 1

(a) $0 \le P(E) \le 1$ **(b)** $P(E) > 1$ **(c)** $P(E) < 0$ **(d)** $-1 \le P(E) \le 1$

6. A sequence in which each term is obtained by adding a fixed number to the previous term is called: 1

(a) a constant sequence **(b)** an arithmetic progression
(c) a Fibonacci sequence **(d)** a geometric progression

7. Which of the following numbers has an odd number of factors? 1

(a) 27 **(b)** 24 **(c)** 25 **(d)** 20

8. The average of 8 numbers is 32. If the number 39 is removed, the average of the remaining 7 numbers is: 1

(a) 33 **(b)** 32 **(c)** 30 **(d)** 31

9. By Euclid's algorithm, gcd$(138, 117)$ is: 1

(a) 2 **(b)** 3 **(c)** 5 **(d)** 4

10. The diagonal of a square is 8 cm. Its area is: 1

(a) 48 cm^2 **(b)** 36 cm^2 **(c)** 32 cm^2 **(d)** 40 cm^2

11. The slope of the line $y = 5$ is: 1

(a) 1 **(b)** undefined **(c)** 5 **(d)** 0

12. The volume of a pyramid with base area B and height h is: 1

(a) $\frac{1}{3}Bh$ **(b)** $\frac{2}{3}Bh$ **(c)** Bh **(d)** $\frac{1}{2}Bh$

13. The distance between $(-1, -1)$ and $(4, 11)$ is: 1

(a) 15 **(b)** 13 **(c)** 14 **(d)** 12

14. Which of these is a linear polynomial? 1

(a) 9 **(b)** $x^2 - 5$ **(c)** x^3 **(d)** $5 - 2x$

15. $x = 0.\overline{45}$ equals: 1

(a) $\frac{5}{12}$ **(b)** $\frac{4}{11}$ **(c)** $\frac{5}{11}$ **(d)** $\frac{6}{11}$

16. $(3x + 4y)^2 =$ 1

(a) $9x^2 + 16y^2$ **(b)** $9x^2 - 24xy + 16y^2$ **(c)** $9x^2 + 12xy + 16y^2$ **(d)** $9x^2 + 24xy + 16y^2$

17. The radius of a circle is 25 cm and a chord is 30 cm long. The distance of the chord from the centre (in cm) is: 1

(a) 21 **(b)** 22 **(c)** 20 **(d)** 19

18. The area of a triangle with sides 7 cm, 15 cm and 20 cm is (cm^2): 1

(a) 41 **(b)** 42 **(c)** 43 **(d)** 44

19. A die is thrown 100 times with frequencies of $1, \ldots, 6$ being 15, 22, 17, 16, 14, 16. The experimental probability of getting 5 is: 1

(a) $\frac{43}{50}$ **(b)** $\frac{4}{25}$ **(c)** $\frac{7}{50}$ **(d)** $\frac{7}{51}$

20. If $t_n = 3n^2 - 4$, then t_4 is: 1

(a) 43 **(b)** 44 **(c)** 45 **(d)** 46

21. The converse of the proposition “If a quadrilateral is a square, then it is a rectangle” is: 1

(a) If a quadrilateral is a square, then it is a rectangle.
(b) If a quadrilateral is a rectangle, then it is a square.
(c) If it is not true that a quadrilateral is a square, then it is not true that it is a rectangle.
(d) If it is not true that it is a rectangle, then a quadrilateral is a square.

22. The average marks of 35 students is 65 and of another 55 students is 47. The average of all 90 students is: 1

(a) 54 **(b)** 53 **(c)** 55 **(d)** 56

23. Adding 1497 and 3340 by the column algorithm, the number of columns in which a carry is generated is: 1

(a) 3 **(b)** 0 **(c)** 2 **(d)** 1

24. The angles of a quadrilateral are in the ratio $2 : 6 : 1 : 3$. The largest angle is: 1

(a) 190° **(b)** 200° **(c)** 180° **(d)** 170°

25. Is the point $(5, -6)$ a solution of $6x + 4y = 6$? 1

(a) No **(b)** Cannot be said **(c)** Only for $x = 0$ **(d)** Yes

26. Take $\pi = \frac{22}{7}$. The curved surface area of a cylinder of radius 28 cm and height 27 cm is: 1

(a) 5228 cm^2 **(b)** 4276 cm^2 **(c)** 4752 cm^2 **(d)** 5704 cm^2

27. The midpoint of the segment joining $(-6, -2)$ and $(2, 10)$ is: 1

(a) $(-2, 4)$ **(b)** $(4, -2)$ **(c)** $(-2, -4)$ **(d)** $(2, 4)$

MASTERMIND

28. If $x = 4$ is a zero of the polynomial $3x + k$, then $k =$ 1

(a) -10 **(b)** -13 **(c)** -11 **(d)** -12

29. An irrational number between 3 and 4 is: 1

(a) $\sqrt{17}$ **(b)** $\sqrt{18}$ **(c)** $\sqrt{7}$ **(d)** $\sqrt{11}$

30. $(2x - 5y)^2 =$ 1

(a) $4x^2 + 25y^2$ **(b)** $4x^2 - 10xy + 25y^2$ **(c)** $4x^2 + 20xy + 25y^2$ **(d)** $4x^2 - 20xy + 25y^2$

31. The chord AB subtends 88° at the centre O. The angle subtended at a point on the major arc is: 1

(a) 64° **(b)** 44° **(c)** 34° **(d)** 54°

32. The semi-perimeter of a triangle with sides 9 cm, 10 cm and 17 cm is (cm): 1

(a) 17 **(b)** 19 **(c)** 20 **(d)** 18

33. In a survey of 400 families, 90 owned a two-wheeler. The probability that a family chosen at random owns a two-wheeler is: 1

(a) $\frac{9}{41}$ **(b)** $\frac{9}{40}$ **(c)** $\frac{31}{40}$ **(d)** $\frac{1}{4}$

34. Which term of the sequence $t_n = 4n - 3$ is 81? 1

(a) 22th term **(b)** 23th term **(c)** 21th term **(d)** 20th term

35. The proposition “If a quadrilateral is a square, then it is a rectangle” is true. Its converse “If it is a rectangle, then a quadrilateral is a square” is: 1

(a) It cannot be decided **(b)** False

(c) True **(d)** True only for some values

36. The average of 21 numbers is 31. When a number 53 is included, the average of all 22 numbers becomes: 1

(a) 33 **(b)** 32 **(c)** 31 **(d)** 34

37. The divisor algorithm applied to 77 produces a list with how many entries? 1

(a) 4 **(b)** 3 **(c)** 5 **(d)** 6

38. In a parallelogram, two opposite angles are $(3x + 10)^\circ$ and $(x + 68)^\circ$. The value of x is: 1

(a) 29 **(b)** 28 **(c)** 31 **(d)** 30

39. The x-intercept of the line $5x + 4y = 59$ is: 1

(a) $\frac{64}{5}$ **(b)** $\frac{59}{5}$ **(c)** $\frac{54}{5}$ **(d)** $\frac{69}{5}$

40. Take $\pi = \frac{22}{7}$. The total surface area of a cylinder of radius 10.5 cm and height 18 cm is: 1

(a) 2259 cm^2 **(b)** 1692 cm^2 **(c)** 1881 cm^2 **(d)** 2070 cm^2

41. A point on the x-axis at distance 4 units to the left of the origin has coordinates: 1

(a) $(4, 0)$ **(b)** $(-4, 0)$ **(c)** $(-3, 0)$ **(d)** $(0, -4)$

42. The value of $2x + 7$ at $x = -4$ is: 1

(a) -2 **(b)** -1 **(c)** 0 **(d)** 1

43. The product of a non-zero rational number and an irrational number is: 1

(a) an integer **(b)** irrational **(c)** zero **(d)** rational

44. $(x + 8)(x - 5) =$ 1

(a) $x^2 + 13x - 40$ **(b)** $x^2 + 3x - 40$ **(c)** $x^2 + 3x + 40$ **(d)** $x^2 - 3x - 40$

MASTERMIND

45. The chord AB subtends 98° at the centre. The angle it subtends at a point on the minor arc is: 1

(a) 141° **(b)** 151° **(c)** 131° **(d)** 121°

46. The area of a circle with circumference 132 cm (take $\pi = \frac{22}{7}$) is (cm^2): 1

(a) 1385 **(b)** 1386 **(c)** 1387 **(d)** 1388

47. A letter is chosen at random from the word RAJASTHAN. The probability that it is a vowel is: 1

(a) $\frac{2}{3}$ **(b)** $\frac{1}{6}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{1}{3}$

48. A sequence has $t_1 = 9$ and $t_{n+1} = t_n + 4$. Then t_6 equals: 1

(a) 28 **(b)** 29 **(c)** 30 **(d)** 31

49. Which of the following is a counterexample to the statement "$n^2 + n + 41$ is prime for every natural number n"? 1

(a) $n = 40$ **(b)** $n = 5$ **(c)** $n = 10$ **(d)** $n = 20$

50. Marks of 93, 80 and 57 are given weights 3, 2 and 6 respectively. The weighted average is: 1

(a) 73 **(b)** 71 **(c)** 72 **(d)** 70

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND